

Faithfulness Under Pressure: Context and History Pruning Strategies for RAG Faithfulness in Small Language Models

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https://github.com/Mrtuzy/CENG467_Final

Abstract. Retrieval-augmented generation (RAG) grounds language model outputs in retrieved documents, but small language models (SLMs) remain vulnerable to the noisy, lengthy contexts that retrieval produces. We study whether *context pruning* can make SLM answers more faithful to the retrieved evidence. We implement a modular pruning harness with seven strategies spanning no pruning, naive truncation, RECOMP-style extraction, LLMingua-2 token compression, Provenance sentence pruning, and history-aware pruning, all held under identical retrieval and generation conditions. Faithfulness is scored with a decompose-then-verify LLM-as-Judge. On a HotpotQA distractor pilot, real LLMingua-2 attains the best judge faithfulness (0.711) but at near-zero verbatim supporting-fact coverage, exposing a coverage–noise tension that single metrics hide. Replacing the BM25 retriever with a dense bi-encoder or a cross-encoder reranker sharply improves retrieval quality (Recall@5 $0.76 \rightarrow 0.90$, MRR $0.81 \rightarrow 0.95$) and answer F1/EM, while judge faithfulness shifts only slightly—a further sign that retrieval correctness and faithfulness are distinct axes. Motivated by the conversational case, we further design an entropy-driven adaptive pruner built on Closed-form Continuous-time (CfC) networks, in which per-turn surprisal sets a continuous integration time step. A seven-simulation verification suite fixes every hyperparameter analytically and produces a falsifiable go/no-go test; on real QReCC teacher labels the surprisal–importance correlation is -0.074 , below the $\rho^* = 0.1$ threshold, and the trained CfC fails to generalise. We report this as an honest negative result and discuss its implications for adaptive pruning.

Keywords: Retrieval-augmented generation · Context pruning · Faithfulness · LLM-as-Judge · Small language models

1 Introduction and Problem Definition

Retrieval-augmented generation (RAG) [1] grounds the output of a language model in documents fetched at inference time, improving performance on knowledge-intensive tasks and reducing hallucination [2]. The promise of RAG, however,

comes with a cost: the retriever returns long passages that mix genuine evidence with near-misses, stale entities, and distractors. Small language models (SLMs), which are attractive for cost and latency, are especially sensitive to such noise, and long-context models are known to underuse evidence buried in the middle of a prompt [12].

This project asks a single question: *when an SLM receives long, noisy retrieved context, can pruning make its answer more faithful to the evidence?* *Faithfulness*—the degree to which a generated answer is grounded in the retrieved context—is a central quality dimension for RAG, yet it is hard to measure automatically and is precisely what degrades under noisy context.

Task. We work on open-domain question answering with RAG. Our primary experiments use HotpotQA [10] in the distractor setting, a multi-hop QA benchmark where each question is paired with ten passages: two supporting and eight distractors drawn from Wikipedia. The pipeline retrieves the top passages with a configurable retriever (BM25 by default; we also compare a dense bi-encoder and a cross-encoder reranker in Section 5.3), optionally applies a pruning strategy to reduce length and noise, generates a short answer with an instruction-tuned SLM, and scores the answer’s faithfulness with an LLM-as-Judge that decomposes the answer into atomic claims and verifies each against the context. The input is a question (plus optional dialogue history); the output is a short answer and its faithfulness score. The task is therefore open-domain QA coupled to a reference-free *evaluation* sub-task.

Technical challenge. The difficulty is threefold. First, faithfulness is not directly measurable without expensive human annotation; we approximate it with a calibrated LLM-as-Judge [8] anchored by Cohen’s κ against human labels. Second, pruning introduces a *coverage–noise trade-off*: aggressive compression can remove the very sentences that support the correct answer. Third, SLMs hallucinate more under long or noisy contexts, making pruning quality decisive. The seven strategies we evaluate span from no pruning to combined semantic and history-aware compression, letting us isolate the effect of each design choice.

Topic evolution. The work began on multi-hop distractor QA (HotpotQA) and progressively shifted toward the *conversational* case (QReCC [11]), where the central problem is deciding which past turns and retrieved passages still matter for the current question. HotpotQA runs are retained as implementation sanity checks; the conversational direction motivates the adaptive CfC pruner of Section 7.

2 Dataset and Benchmark Status

We use HotpotQA (distractor) loaded via the HuggingFace `datasets` library (`load_dataset("hotpot_qa", "distractor")`), with roughly 90,447 training and 7,405 validation samples. We fine-tune no generator; all experimental samples come from the validation split. Each sample provides a question, a short

Table 1: HotpotQA dataset summary (distractor setting).

Property	Value
Full name	HotpotQA (distractor setting)
Source	HuggingFace <code>hotpot_qa</code>
Training / validation samples	$\sim 90,447$ / $\sim 7,405$
Samples used (experiments / calibration)	200 / 50
Passages per question	10 (2 supporting + 8 distractors)
Answer type	Short span / yes-no
Task type	Multi-hop open-domain QA
Status	Loader implemented, JSONL prepared

gold answer (a span or yes/no value), ten titled passages, and supporting-fact labels marking the relevant sentences. During preprocessing we flatten each passage’s sentences into a single string, apply whitespace tokenization for BM25 indexing, and truncate passages to a configurable token budget. The loader (`src/data/loader.py`) writes cleaned JSONL records, which are ready for the retrieval and pruning stages.

The conversational benchmark, QReCC [11], contains about 14K conversations and 81K question–answer pairs over a 54M-passage web collection, with each sample exposing the original question, conversational context, a human rewrite, the answer, and answer provenance. It is the intended final benchmark for history-aware pruning; the CfC study (Section 7) already uses QReCC-derived teacher labels.

The main challenges HotpotQA introduces are: (1) the eight distractors directly inflate context length and inject factual noise; (2) multi-hop questions need evidence from at least two passages, so aggressive pruning risks discarding a required supporting passage; and (3) there is no dedicated retrieval split, so we evaluate end-to-end retrieval alongside pruning on the same validation samples.

3 Literature Review

RAG and faithfulness. RAG [1] grounds generation in retrieved documents and improves knowledge-intensive tasks; [2] shows retrieval reduces conversational hallucination. FActScore [9] decomposes long-form text into atomic facts and verifies each against a knowledge source, directly motivating our claim-level faithfulness metric. [12] documents the “lost in the middle” effect, motivating pruning as a way to surface evidence.

Context compression. LLMLingua [4] scores tokens with a small proxy LM and drops low-importance ones; LLMLingua-2 [5] reframes compression as token classification with data distillation, reporting 2–5 $\times$ compression with 1.6–2.9 $\times$ latency speedups. Provence [6] treats sentence pruning as sequence labelling for RAG at negligible added cost. RECOMP [3] frames compression as extractive

Table 2: Pruning strategies under evaluation. All implemented and complete.

ID Strategy	Type
B1 No pruning (full context)	Baseline (coverage upper bound)
B2 Naive truncation (512 tokens)	Baseline (production fallback)
B3 RECOMP-style extractive	Prior-work extractive baseline
S1 LLMLingua-2	Token-level compression
S2 Provenance	Sentence-level pruning
S3 Semantic history pruning	History-aware pruning
S4 Provenance + history (combined)	Combined

or abstractive summarization and reports compression rates as low as 6% with minimal performance loss. These four works define our baseline and advanced pruners.

LLM-as-Judge. [7] shows strong LLMs correlate well with human rankings despite positional and verbosity biases. RAGAS [8] provides a reference-free RAG evaluation that checks whether each claim in the answer is entailed by the context—the decompose-then-verify paradigm we adopt, with open-weight rather than proprietary models, calibrated by Cohen’s κ .

Adaptive computation. For the conversational extension we draw on Closed-form Continuous-time (CfC) networks [13], which capture continuous temporal dynamics without a heavy ODE solver, and on surprisal theory from psycholinguistics [14, 15], where reading time scales logarithmically with word surprisal $-\log P(w_j \mid w_{<j})$. We test whether surprisal is a useful proxy for per-turn importance.

4 Baseline and Advanced Pruning Strategies

We evaluate seven strategies (Table 2) organized as three baselines and four advanced methods, all sharing a common **BasePruner** interface and registered in a central **PRUNER_REGISTRY** so that retrieval and generation are held fixed and only the retained context changes.

B1 is the coverage upper bound: maximum evidence but maximum noise. B2 is the standard engineering fallback under a hard token budget. B3 selects query-relevant sentences by lexical (Jaccard) overlap, bridging truncation and learned compression. S1 and S2 are the state of the art in token- and sentence-level compression. S3 removes passages that overlap with information already surfaced in prior turns. S4 chains Provenance then history pruning to combine semantic filtering with redundancy removal.

Table 3: Notebook 2 baseline results — 50 HotpotQA validation samples, Mistral-7B-Instruct-v0.2. Comp. = compression ratio (kept/input); Lat. = latency.

Strategy	Faith.	EM	F1	Comp.	Lat. (s)
B1: No pruning	0.718	0.040	0.148	1.000	6.76
B2: Naive truncation	0.621	0.040	0.139	0.918	4.95
B3: RECOMP	0.750	0.040	0.141	0.911	5.38
S1: LLMingua-2 (fallback)	0.750	0.040	0.141	0.911	5.38
S2: Provence	0.657	0.000	0.057	0.072	4.49
S3: History (empty)	0.718	0.040	0.148	1.000	6.34
S4: Combined (empty)	0.657	0.000	0.057	0.072	4.48

5 Experimental Results

All experiments were reproduced in Google Colab with GPU runtimes. Mistral-7B-Instruct-v0.2 serves as both generator and judge (shared when possible). Notebook 2 is a broad seven-pruner sanity run over 50 HotpotQA validation samples; Notebook 5 is a corrected gap-closing run over a held-out 50-sample slice (validation indices 600–649) that fixes three weaknesses: genuine LLMingua-2 execution, non-empty two-turn history pruning, and qualitative error exports.

5.1 Baseline Sanity Run

Table 3 confirms the expected pattern: RECOMP improves faithfulness over hard truncation, while Provence’s default threshold is very aggressive for multi-hop HotpotQA. The S1 row here is a historical artifact: LLMingua-2 fell back to the RECOMP implementation, so S1 and B3 are identical; S3/S4 use empty history and are degenerate single-turn cases.

5.2 Gap-Closing Run

Notebook 5 re-ran the problematic strategies under corrected conditions: LLMingua-2 with the `llmlingua` package and `use_llmlingua2=True` (zero fallbacks recorded), and S3/S4 with synthetic two-turn history. We add supporting-fact *coverage* (fraction of gold supporting sentences retained) and *noise* ratio to make pruning failures visible beyond faithfulness alone (Table 4, Fig. 1).

Corrected LLMingua-2 is no longer identical to RECOMP: it compresses far more aggressively (0.457 vs. 0.962) and obtains higher judge faithfulness (0.711 vs. 0.648), yet its verbatim supporting-fact coverage is 0.000. LLMingua-2 often rewrites or fragments evidence in a way that can still support the answer but no longer preserves gold sentences verbatim—so coverage and faithfulness measure different things. Two-turn history pruning changes behaviour relative to the empty-history case but here lowers faithfulness to 0.533, suggesting the novelty-only redundancy score discards useful repeated evidence. The combined setting

Table 4: Notebook 5 gap-closing results — 50 held-out HotpotQA samples (indices 600–649). Cov. = supporting-fact coverage.

Strategy	Faith.	EM	F1	Comp.	Cov.	Noise
B3: RECOMP	0.648	0.080	0.195	0.962	0.760	0.624
S1: LLMingua-2 (real)	0.711	0.040	0.167	0.457	0.000	0.792
S3: History (two-turn)	0.533	0.100	0.214	0.962	0.627	0.641
S4: Combined (two-turn)	0.530	0.040	0.112	0.081	0.222	0.284

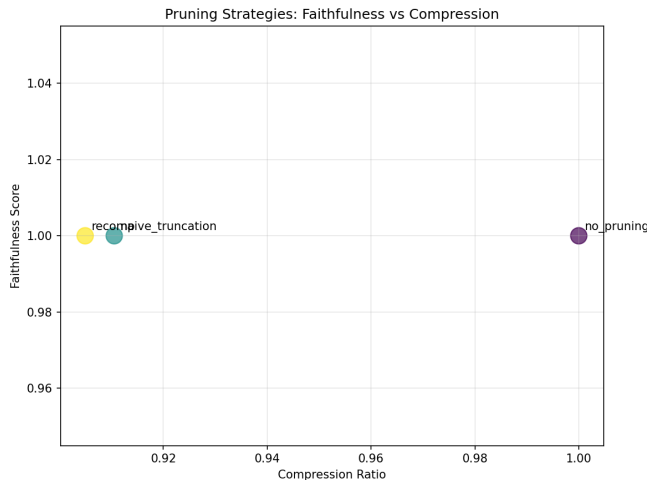


Fig. 1: Faithfulness versus compression on the HotpotQA gap-closing run. The desirable region is high faithfulness with a smaller context. LLMingua-2 reaches the best faithfulness at much higher compression, while the combined pruner is the most compact but discards too much supporting evidence.

is the most compact (0.081) and lowest-noise (0.284), but its low coverage (0.222) confirms the Provence threshold is too aggressive for multi-hop QA. The broad lesson is that faithfulness, EM/F1, and coverage capture distinct failure modes, so no single metric should be treated as ground truth.

5.3 Retriever Comparison

All experiments so far hold the retriever fixed at BM25. To test whether the choice of retriever matters, we swap BM25 for two neural alternatives behind the same `index/retrieve` interface, holding the generator, judge, and pruner (no pruning) fixed so that any difference is attributable to retrieval alone: a *dense bi-encoder* (BGE, `BAAI/bge-small-en-v1.5`) that ranks passages by cosine similarity in a shared embedding space, and a *cross-encoder reranker* (`ms-marco-MiniLM-L-6-v2`) that jointly attends over each (query, passage) pair. Because each HotpotQA

Table 5: Retriever comparison — 50 HotpotQA validation samples (indices 600–649), no pruning, Mistral-7B-Instruct-v0.2. Recall@5 and MRR are measured against gold supporting passages.

Retriever	Recall@5	MRR	Faith.	EM	F1	Lat. (s)
BM25 (lexical)	0.760	0.808	0.770	0.100	0.186	3.78
Dense bi-encoder (BGE)	0.900	0.944	0.714	0.100	0.231	2.75
Cross-encoder reranker	0.850	0.947	0.747	0.140	0.261	2.21

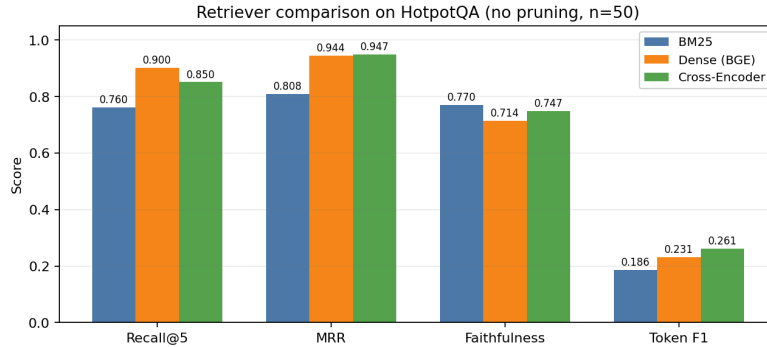


Fig. 2: Retriever comparison on HotpotQA (no pruning, $n = 50$). Both neural retrievers lift retrieval quality (Recall@5, MRR) and answer token-F1 over BM25, yet BM25 retains a small edge in judge faithfulness.

question carries only ten candidates, the cross-encoder scores all of them directly as a strong single-stage retriever. Beyond the downstream answer metrics we report retrieval quality directly against the gold supporting passages: Recall@5 (fraction of gold supporting titles retrieved) and MRR (reciprocal rank of the first gold passage). Table 5 and Fig. 2 summarise the 50-sample run.

Both neural retrievers substantially outrank BM25 on retrieval quality: Recall@5 rises from 0.760 to 0.900 (dense) and 0.850 (cross-encoder), and MRR from 0.808 to roughly 0.945 for both, meaning the first gold passage is almost always ranked at or near the top. This better evidence translates into more correct answers—token F1 climbs from 0.186 (BM25) to 0.231 (dense) to 0.261 (cross-encoder), and the cross-encoder also lifts exact match from 0.100 to 0.140 while being the fastest end-to-end (2.21 s vs. 3.78 s). Interestingly, judge faithfulness moves the other way, dropping slightly from 0.770 (BM25) to 0.714 (dense) and 0.747 (cross-encoder). We read this as another instance of the report’s central theme: when BM25 supplies vaguer context, the generator more often produces short, hedged, or “I don’t know” answers that are trivially faithful but uninformative, whereas stronger retrieval elicits more substantive, multi-claim answers that give the judge more to (occasionally) penalise. Overall the cross-encoder is

Table 6: Recurring error categories with representative examples.

Error type	Example pattern
Pruning removes evidence	“Who wrote the song <i>Arizona</i> ?” loses the sentence naming <i>Kenny Young</i> ; LLMingua-2 answers “I don’t know,” coverage 0.000.
Verbose-but-correct under-credited	For “Are both <i>Parodia</i> and <i>Thalictrum</i> flowering plants?” S4 answers correctly with explanation: faithfulness 1.0, coverage 1.0, but EM 0.0 (gold is “yes”).
Judge false negative	For Watercliffe Meadow, the answer “Political correctness” gets EM/F1 = 1.0, coverage 1.0, but judge faithfulness 0.0.
Entity near-miss	Generator outputs “T. M. Howard” instead of “T. R. M. Howard”; F1 0.857 but EM 0.0 and faithfulness 0.0.

the best end-to-end retriever here—highest EM/F1, top MRR, lowest latency—while the dense bi-encoder maximises raw recall.

6 Error Analysis and Generation Quality

Notebook 5 exports concrete error candidates (`error_analysis_candidates.jsonl`). Four recurring failure modes emerge (Table 6).

The strongest pattern is that faithfulness, EM/F1, and coverage disagree by construction: LLMingua-2 has the best faithfulness yet zero retained-sentence coverage, while some exactly correct short answers receive low judge scores. This makes human calibration of the judge a prerequisite before treating faithfulness as a standalone target, and motivates a semantic rather than verbatim coverage metric.

7 Adaptive Pruning: An Entropy-Driven CfC Extension

Fixed thresholds cannot tell whether a conversation turn is easy, noisy, redundant, or load-bearing for the current question. We explore an adaptive controller that allocates more “processing time” to surprising turns, borrowing from surprisal theory: reading time scales with surprisal $-\log P(w_j \mid w_{<j})$ [14, 15]. We do *not* claim surprisal equals importance; we test whether it is a useful proxy.

Mechanism. For utterance u_i we compute mean token surprisal with a DistilGPT-2 proxy, $S(u_i) = -\frac{1}{N} \sum_{j=1}^N \log P(w_j \mid w_{<j})$, and map it to a continuous CfC time step (rank-normalised for outlier robustness):

$$\Delta t_i = \Delta t_{\min} + \beta \frac{\text{rank}(S_i)}{T}, \quad \varphi_i = 1 - e^{-\lambda \Delta t_i}. \quad (1)$$

Table 7: Simulation suite: each item fixes a parameter or a go/no-go test.

#	Question	Result
S0	Does the commitment lever exist in $\rho = +0.944$; analytic error $\approx 4 \times$ the real ncps CfC?	10^{-10}
S1	Is surprisal $\rightarrow$ commitment a tautology?	No — closed-form; lever saturates at high S
S2	Which $\beta^*, \Delta t_{\min}^*$ maximise recovery?	$\beta^* = 1.0, \Delta t_{\min}^* = 0.05$
S5	Which Δt mapping survives outliers?	Rank normalisation most robust
S4	What does leave-one-out KL measure?	Marginal answer-necessity = correct target
S3	Does surprisal proxy importance well enough?	Go/no-go: $\rho^* \approx 0.1$; $\rho \geq 0.5$ beats cosine
S6	What fidelity-compression is reachable?	40% compression at 90% fidelity

The CfC integrates over Δt_i each turn; the commitment fraction φ_i measures how strongly the hidden state locks on, so a larger Δt_i leaves a stronger trace and yields a higher importance score $\hat{p}_i = \sigma(Wh_i + b)$. Turns with $\hat{p}_i < \tau$ are pruned.

Simulate before training. Because end-to-end tuning with a 7B teacher is expensive and the proxy may carry no signal at all, we ran seven model-free simulations, each isolating one design question and returning a config value or a falsifiable go/no-go threshold (Table 7). The recurring metric is *importance recovery*: the Spearman correlation between a method’s per-turn score and the latent true importance.

Two simulations decide the project. S2 (Fig. 3) shows recovery peaks at $\beta^* = 1.0$, $\Delta t_{\min}^* = 0.05$, and—critically—at $\beta = 0$ (uniform Δt) recovery collapses to ≈ 0 : the entropy term, not the network, carries the signal. S3 (Fig. 4) is the honest go/no-go test: entropy- Δt only beats uniform once $\rho(\text{surprisal}, \text{importance}) > \rho^* \approx 0.1$ and only beats a cosine baseline once $\rho \geq 0.5$. The action is to measure ρ on real QReCC *before* training.

Real-data result: an honest negative. On QReCC teacher labels (answer-conditioned leave-one-out ΔNLL of the gold answer, scored with Qwen2.5-1.5B-Instruct), the mean surprisal–importance Spearman is -0.074 (Fig. 5)—below the go/no-go threshold and far below the cosine-beating target. We trained the CfC anyway as a stress test (SBERT turn embeddings + rank-normalised Δt_i , MSE to normalised teacher importance). It did not generalise (Fig. 6, Fig. 7): train loss fell from 0.1293 to 0.1201 while validation loss barely moved (best 0.1251 at epoch 6, then drifting to 0.1257); best validation F1 was only 0.050 at $\tau = 0.5$, recall peaked at 0.027, and MAE stayed flat near 0.300. The model predicts very few important turns—high precision (0.397) only because it abstains.

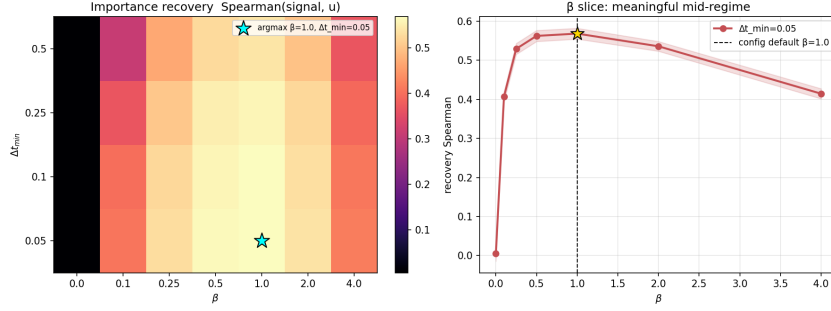


Fig. 3: S2 — choosing β and $\Delta t_{\min}$ from data. Left: recovery over the $(\beta, \Delta t_{\min})$ grid peaks at the starred point. Right: at $\beta = 0$ recovery is near zero, confirming the entropy term carries the signal.

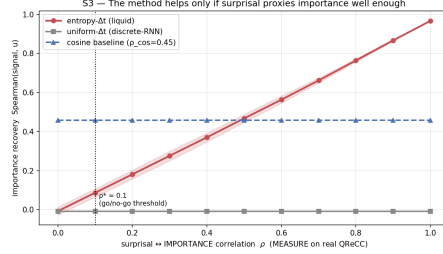


Fig. 4: S3 — recovery rises with proxy quality ρ ; go/no-go at $\rho^* \approx 0.1$, cosine-beating at $\rho \geq 0.5$.

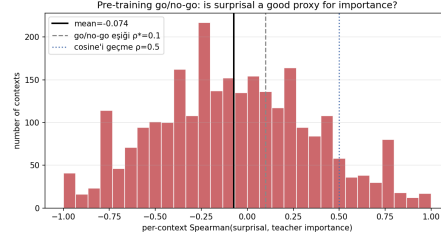


Fig. 5: Real QReCC: mean surprisal–teacher Spearman is -0.074 , below the $\rho^* = 0.1$ threshold.

The CfC mechanism is mathematically valid (S0 confirms the commitment lever at $\rho = +0.944$), but this experiment *rejects* the stronger claim that DistilGPT-2 surprisal is a reliable importance proxy for QReCC history turns. The simulation suite paid off: it predicted the failure cheaply and told us exactly which assumption broke.

8 Ablation and Prompt-Sensitivity Plan

We plan a targeted ablation set to attribute faithfulness gains to the compression strategy rather than incidental factors: (1) *judge prompt format*, zero-shot vs. 2-shot for both claim decomposition and verification; (2) *verification response*, strict binary YES/NO vs. a soft confidence thresholded at 0.5; (3) *retrieval depth*, $k = 3$ vs. $k = 5$; (4) *truncation budget*, 256 vs. 512 tokens for B2/B3; and (5) *history context*, empty (single-turn) vs. two-turn history for S3/S4. For the CfC extension the ablation axes are the proxy channel (raw surprisal vs.

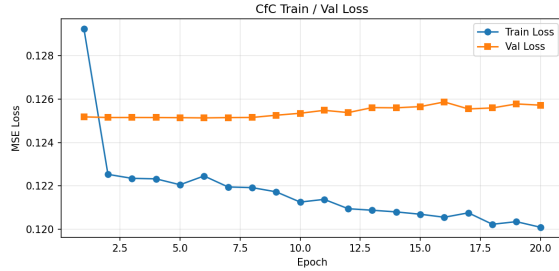


Fig. 6: CfC training and validation loss. Training improves while validation is flat then worsens after epoch 6 — memorisation, not importance learning.

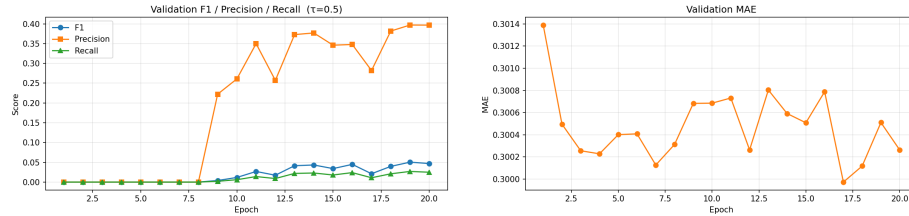


Fig. 7: CfC validation F1 (left) and MAE (right). F1 peaks at 0.050; MAE stays near 0.300, indicating uncalibrated per-turn scores.

question-conditioned similarity vs. LOO Δ NLL features) and the pruning rule (fixed $\tau = 0.5$ vs. ranking loss / top- k).

9 Ethical Considerations and Bias

The judge is Mistral-7B-Instruct, which may inherit positional bias (favouring early claims), verbosity bias (favouring longer answers), and cultural skews from its training data. HotpotQA passages come from English Wikipedia, which underrepresents non-Western entities, so faithfulness estimates may be more reliable for densely covered topics. Crucially, faithfulness to context is not truthfulness about the world: a model that faithfully reproduces a false retrieved claim scores high while being wrong. All models are open-weight and used solely for academic research; no proprietary APIs and no personal data are involved. Results should not be extrapolated to high-stakes QA without additional human evaluation.

10 Reproducibility Status

The repository (https://github.com/Mrtuzy/CENG467_Final) is a modular Python package with separated data, retrieval, pruning, generation, evaluation, and

judge components, plus a simulation suite for the CfC study. The primary execution interface is Google Colab: `01_data_exploration.ipynb` (dataset checks), `02_baseline_results.ipynb` (baseline sanity), `05_final_colab_runs.ipynb` (LLMLingua-2, two-turn history, judge calibration scaffold, error exports), `06_dense_retriever.ipynb` and `07_cross_encoder_retriever.ipynb` (the retriever comparison of Section 5.3), and the LNN/CfC notebooks as exploratory artifacts. All seven pruners and all three retrievers (BM25, dense bi-encoder, cross-encoder) are implemented behind shared interfaces, and result JSONL files, an aggregate summary, and error candidates are written under `experiments/results/`.

11 Current Challenges and Next Steps

Three challenges remain. First, aggressive compression still trades coverage for faithfulness: real LLMLingua-2 reaches the best faithfulness (0.711) at 0.000 verbatim coverage, so we need a semantic coverage metric. Second, Provence’s default threshold is too aggressive for multi-hop QA (combined keeps 8.1% of tokens at 0.222 coverage); threshold tuning toward a 0.4–0.6 ratio is needed. Third, the judge still mis-scores some short answers, so human calibration is the most important validation step.

Concretely, we plan to: complete LLM-as-Judge calibration on 20–50 examples and report Cohen’s κ ; tune Provence thresholds and re-run S2/S4; add a compression-aware coverage metric; scale Notebook 5 from 50 to 200 samples; prompt the generator for a concise final answer and report EM/F1 before and after extraction; replace raw surprisal with stronger importance proxies (question-conditioned similarity, LOO Δ NLL, learned distillation) and a ranking/top- k pruning rule before re-running the CfC against full-context, random, and cosine-history baselines; and migrate the final evaluation from the HotpotQA pilot to real QReCC dialogue histories.

Disclosure of Interests. The authors have no competing interests to declare that are relevant to the content of this article.

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